INTRODUCTION TO ASYMMETRIC (PUBLIC KEY) ENCRYPTION

Recall: Symmetric-Key Setting

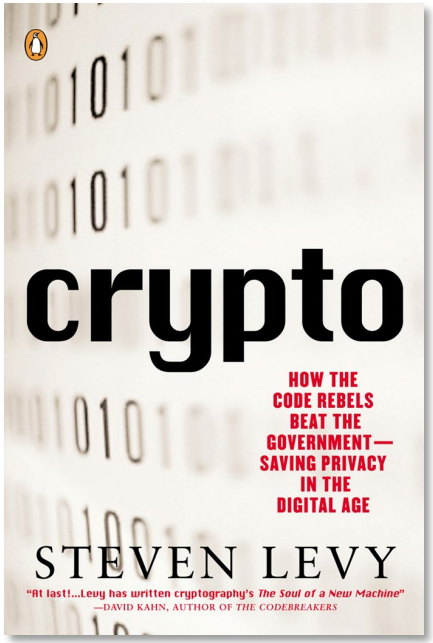


Public-Key (Asymmetric-Key) Setting



Recommended Book

A non-technical account of the history of public-key cryptography and the colorful characters involved:

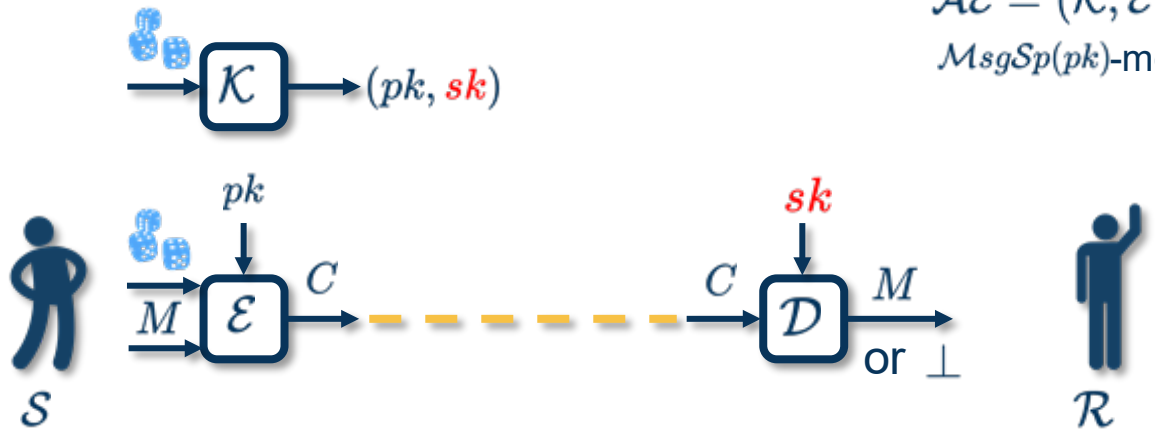


Asymmetric Encryption Scheme

A scheme $\mathcal{AE}$ is specified by a key generation algorithm $\mathcal{K}$, an encryption algorithm $\mathcal{E}$, and a decryption algorithm $\mathcal{D}$.

$\mathcal{AE} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$

$\mathcal{MsgSp}(pk)$-message space



It is required that for every (pk, sk) that can be output by $\mathcal{K}$ and every $M \in \mathcal{MsgSp}(pk)$ if $C = \mathcal{E}(pk, M)$ then $\mathcal{D}(sk, C) = M$

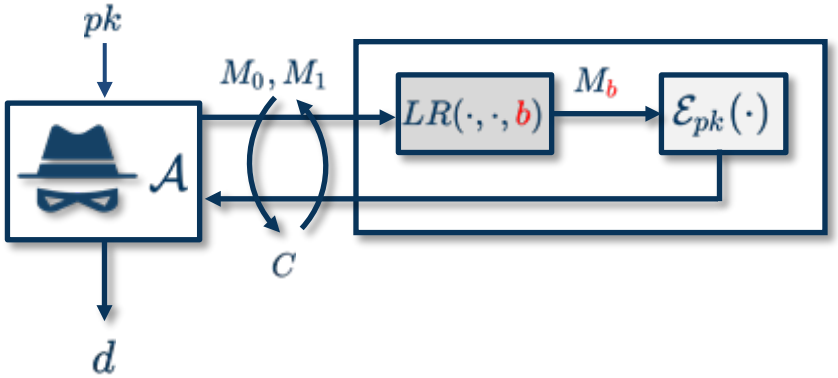
- A sender must know the receiver's public key, and must be assured that this public key is authentic (really belongs to the receiver). This is ensured be the PKI processes, which are not part of encryption.
- Unlike in a symmetric encryption, the encryption algorithm is never stateful.
- What is secure asymmetric encryption scheme?

Indistinguishability under Chosen-Plaintext Attacks

Fix $\mathcal{AE} = (\mathcal{K}, \mathcal{E}, \mathcal{D}), (pk, sk) \xleftarrow{\$} \mathcal{K}$

For an adversary $\mathcal{A}$ and a bit b (either 0 or 1) consider an experiment ind-cpa-b, i.e., consider two experiments, ind-cpa-0 (“left”) and ind-cpa-1 (“right”).

In each experiment, $\mathcal{A}$ is given access to the “left-right” encryption oracle that takes two messages and returns encryption of M_b under pk

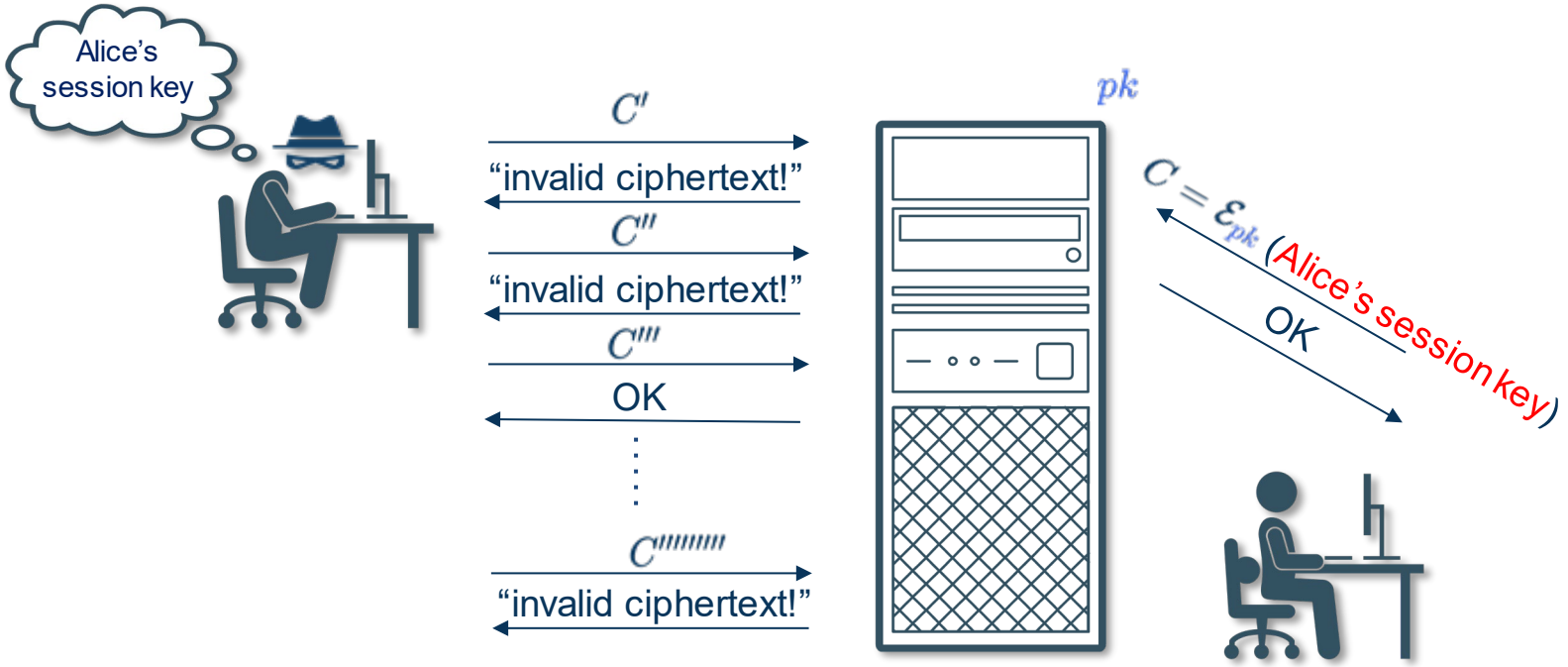


The IND-CPA advantage of $\mathcal{A}$ (denoted as $\mathsf{Adv}^{\text{ind-cpa}}_{\mathcal{AE}}(\mathcal{A})$) is $\Pr[\mathcal{A} \Rightarrow 0$ in ind-cpa-0 exp. $] - \Pr[\mathcal{A} \Rightarrow 0$ in ind-cpa-1 exp. $]$

An asymmetric encryption scheme $\mathcal{AE}$ is indistinguishable under chosen-plaintext attacks (IND-CPA secure) if for any adversary $\mathcal{A}$ with polynomial resources its ind-cpa advantage is negligible in security parameter.

IND-CPA is Not Always Enough

Bleichenbacher's attack on a previous version of SSL:

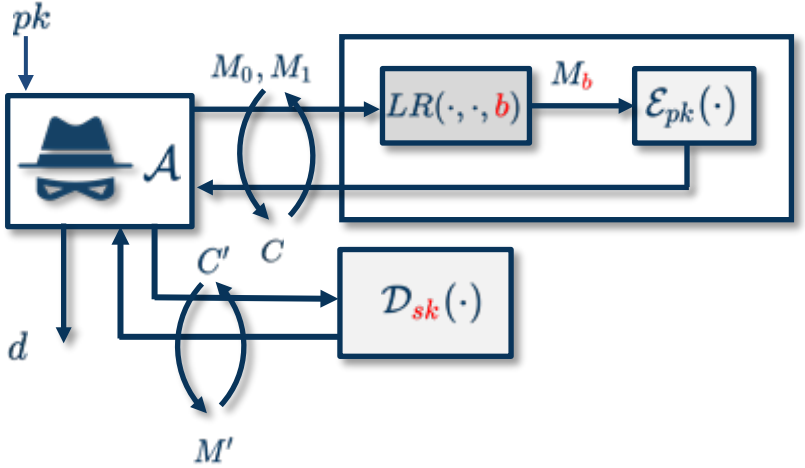


Indistinguishability under Chosen-Ciphertext Attacks

Fix $\mathcal{AE} = (\mathcal{K}, \mathcal{E}, \mathcal{D}), (pk, sk) \xleftarrow{\$} \mathcal{K}$

For an adversary $\mathcal{A}$ and a bit b (either 0 or 1) consider an experiment ind-cca-b, i.e., consider two experiments, ind-cca-0 (“left”) and ind-cca-1 (“right”).

In each experiment, $\mathcal{A}$ is given access to the “left-right” encryption oracle that takes two messages and returns encryption of M_b under pk, and decryption oracle. $\mathcal{A}$ cannot query the decryption oracle on ciphertexts from the “left-right” encryption oracle.



The IND-CCA advantage of $\mathcal{A}$ (denoted as $\mathsf{Adv}^{\text{ind-cca}}_{\mathcal{AE}}(\mathcal{A})$) is $\Pr[\mathcal{A} \Rightarrow 0 \text{ in ind-cca-0 exp. }] - \Pr[\mathcal{A} \Rightarrow 0 \text{ in ind-cca-1 exp. }]$

An asymmetric encryption scheme $\mathcal{AE}$ is indistinguishable under chosen-ciphertext attacks (IND-CCA secure) if for any adversary $\mathcal{A}$ with polynomial resources its ind-cca advantage is negligible in the security parameter.

- IND-CCA ⇒ IND-CPA
- In the literature you can see the definitions where an adversary makes a single query to the LR encryption oracle.
- **Theorem 1**. Let $\mathcal{AE} = (\mathcal{K}, \mathcal{E}, \mathcal{D})$ be an asymmetric encryption scheme. Let $\mathcal{A}$ be an ind-cpa adversary who makes at most q queries to its LR encryption oracle. Then there exists an ind-cpa adversary $\mathcal{A}_1$ with the same running time making at most 1 query to its LR encryption oracle and such that

$$\mathsf{Adv}^{\text{ind-cpa}}_{\mathcal{AE}}(\mathcal{A}) \leq q \cdot \mathsf{Adv}^{\text{ind-cpa}}_{\mathcal{AE}}(\mathcal{A}_1)$$

- **Theorem 2**. Let $\mathcal{A}$ be an ind-cca adversary who makes at most q_e queries to its LR encryption oracle. Then there exists an ind-cca adversary $\mathcal{A}_1$ making at most one query to its LR encryption oracle, the same number of decryption queries and having the same running time such that

$$\mathsf{Adv}^{\text{ind-cca}}_{\mathcal{AE}}(\mathcal{A}) \leq q_e \cdot \mathsf{Adv}^{\text{ind-cca}}_{\mathcal{AE}}(\mathcal{A}_1)$$

Question of Huge Importance

- How to construct a secure public key scheme?
- We could also think about how to build an interactive scheme, called key exchange, where both parties have public/secret keys.
- Some intuition:
  - https://youtu.be/U62S8SchxX4
  - https://youtu.be/YEBfamv-_do

Practical Building Blocks

- Practical building blocks for public key crypto come from (big) numbers and hard problems associated with them
- Let's recall the basics about number theory